

ADVANCED QUANTUM MECHANICS

Exercise 10 - 16.12.2025



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Jordan–Wigner transformation

We consider spinless fermions on a one-dimensional chain with N sites. On each site j we have fermionic annihilation and creation operators c_j and $c_j^\dagger$, satisfying the canonical anticommutation relations

$$\{c_j, c_k^\dagger\} = \delta_{jk}, \quad \{c_j, c_k\} = 0, \quad \{c_j^\dagger, c_k^\dagger\} = 0.$$

We want to represent these fermionic operators as products of Pauli matrices acting on a chain of spin-1/2 degrees of freedom. Recall the Pauli matrices and spin ladder operators on site j ,

$$\sigma_j^x, \quad \sigma_j^y, \quad \sigma_j^z, \quad \sigma_j^\pm = \frac{1}{2}(\sigma_j^x \pm i\sigma_j^y),$$

and the identity operator $\mathbb{1}_j$ on site j . The Jordan–Wigner mapping for an N -site chain, relate fermionic operators c_j , $c_j^\dagger$ to spin operators:

$$c_j = \left(\prod_{k < j} \sigma_k^z \right) \sigma_j^- = (\sigma_1^z) \otimes (\sigma_2^z) \otimes \cdots \otimes (\sigma_{j-1}^z) \otimes (\sigma_j^-) \otimes \underbrace{\mathbb{1} \otimes \cdots \otimes \mathbb{1}}_{j+1, \dots, N}, \quad (1a)$$

$$c_j^\dagger = \left(\prod_{k < j} \sigma_k^z \right) \sigma_j^+ = (\sigma_1^z) \otimes (\sigma_2^z) \otimes \cdots \otimes (\sigma_{j-1}^z) \otimes (\sigma_j^+) \otimes \underbrace{\mathbb{1} \otimes \cdots \otimes \mathbb{1}}_{j+1, \dots, N}. \quad (1b)$$

(i) Specialize to a chain with *three* sites, $j = 1, 2, 3$ showing that

$$\begin{aligned} c_1 &= \sigma_1^- \otimes \mathbb{1}_2 \otimes \mathbb{1}_3, & c_1^\dagger &= \sigma_1^+ \otimes \mathbb{1}_2 \otimes \mathbb{1}_3, \\ c_2 &= \sigma_1^z \otimes \sigma_2^- \otimes \mathbb{1}_3, & c_2^\dagger &= \sigma_1^z \otimes \sigma_2^+ \otimes \mathbb{1}_3, \\ c_3 &= \sigma_1^z \otimes \sigma_2^z \otimes \sigma_3^-, & c_3^\dagger &= \sigma_1^z \otimes \sigma_2^z \otimes \sigma_3^+. \end{aligned}$$

fulfill the fermionic commutation relations.

(ii) Define the local fermionic number operator on site j ,

$$\hat{n}_j = c_j^\dagger c_j.$$

Show that on a single site

$$\hat{n}_j = \frac{1}{2}(\mathbb{1}_j - \sigma_j^z),$$

and write the three-site operators $\hat{n}_1, \hat{n}_2, \hat{n}_3$ as tensor products on the full three-qubit space.

(iii) Consider now a purely local fermionic Hamiltonian with on-site energies $\varepsilon_1, \varepsilon_2, \varepsilon_3$,

$$\hat{H} = \varepsilon_1 c_1^\dagger c_1 + \varepsilon_2 c_2^\dagger c_2 + \varepsilon_3 c_3^\dagger c_3.$$

Use your results above to express $\hat{H}$ *entirely* in terms of Pauli matrices and identity operators acting on the three spins. Write the final answer in the form

$$\hat{H} = A_0 \mathbb{1} \otimes \mathbb{1} \otimes \mathbb{1} + A_1 \sigma_1^z \otimes \mathbb{1} \otimes \mathbb{1} + A_2 \mathbb{1} \otimes \sigma_2^z \otimes \mathbb{1} + A_3 \mathbb{1} \otimes \mathbb{1} \otimes \sigma_3^z,$$

and determine the coefficients A_0, A_1, A_2, A_3 in terms of $\varepsilon_1, \varepsilon_2, \varepsilon_3$.

Bogoliubov transformation of a quadratic bosonic Hamiltonian

Consider a single bosonic mode $a, a^\dagger$ with commutator $[a, a^\dagger] = 1$. We study the quadratic Hamiltonian

$$\hat{H} = \omega a^\dagger a + \frac{\lambda}{2} (a^\dagger a^\dagger + aa), \quad \omega > 0, \lambda \in \mathbb{R}.$$

(i) Assume a Bogoliubov transformation

$$b = u a + v a^\dagger, \quad b^\dagger = u a^\dagger + v a,$$

with real coefficients u, v . Impose the bosonic commutation relation $[b, b^\dagger] = 1$ and show that

$$u^2 - v^2 = 1.$$

(ii) Invert the transformation,

$$a = u b - v b^\dagger, \quad a^\dagger = u b^\dagger - v b,$$

and substitute into $\hat{H}$. Expand all terms and collect powers of $b^\dagger b, bb, b^\dagger b^\dagger$ and the identity.

(iii) Demand that the coefficients of bb and $b^\dagger b^\dagger$ vanish. Use $u^2 - v^2 = 1$ to solve for u and v in terms of ω and λ .

(iv) Show that the Hamiltonian becomes diagonal,

$$\hat{H} = \Omega b^\dagger b + \text{constant},$$

and derive the Bogoliubov excitation frequency

$$\Omega = \sqrt{\omega^2 - \lambda^2}.$$

(v) (State and discuss) the condition under which Ω is real and positive. What is the physical interpretation if $\omega^2 - \lambda^2 < 0$?

(vi) Show that the Bogoliubov vacuum contains a finite population of the original a -excitations by computing $\langle 0_b | a^\dagger a | 0_b \rangle$.

Holstein–Primakoff representation of Pauli matrices

Consider a single spin-1/2 degree of freedom described by Pauli matrices $\sigma^x, \sigma^y, \sigma^z$. Recall the identification with spin operators

$$S^\alpha = \frac{1}{2} \sigma^\alpha, \quad \alpha = x, y, z,$$

and the ladder operators

$$S^\pm = S^x \pm iS^y.$$

Introducing a single bosonic mode with operators $a, a^\dagger$ satisfying $[a, a^\dagger] = 1$, the Holstein–Primakoff transformation for general spin length S is

$$S^z = S - a^\dagger a, \quad S^+ = \sqrt{2S - a^\dagger a} a, \quad S^- = a^\dagger \sqrt{2S - a^\dagger a}.$$

(i) Specialize to spin-1/2, so that $S = 1/2$. Show that the Hilbert-space condition for this transformation to be exact is only when cutting off to states 0 and 1.

(ii) The usefulness of the HP transformation becomes evident in large spin systems. Expand the square root to lowest order in $a^\dagger a$, assuming the state is close to the fully polarized $S^z = +S$ state. Show that

$$S^+ \approx \sqrt{2S} a, \quad S^- \approx \sqrt{2S} a^\dagger, \quad S^z \approx S - a^\dagger a.$$

Briefly explain the physical meaning of this approximation.

- (iii) Consider a two-site Heisenberg interaction $\hat{H} = J \mathbf{S}_1 \cdot \mathbf{S}_2$, which describes the coupling of magnetic dipoles stemming from the electronic spins in neighboring sites in a crystal. Insert the linearized Holstein–Primakoff expressions and keep only quadratic terms in $a_{1,2}, a_{1,2}^\dagger$. The resulting Hamiltonian describes free magnon (spin-wave) excitations.

Hint: The quadratic two-boson Hamiltonian can be diagonalized by introducing symmetric and antisymmetric combinations

$$b_{\pm} = \frac{1}{\sqrt{2}} (a_1 \pm a_2), \quad b_{\pm}^\dagger = \frac{1}{\sqrt{2}} (a_1^\dagger \pm a_2^\dagger).$$

In terms of $b_{\pm}$ the Hamiltonian becomes diagonal in the two normal modes, corresponding to the two magnon branches.